

COMPLEX NUMBERS

Pure

A complex number is the sum of a real number and a multiple of "i" where $i = \sqrt{-1}$

Notation:

Generally use "z" for a complex number (sometimes u, v, w)

e.g. $z = 2 + 3i$
 $\uparrow$ $\uparrow$
 real imaginary

$$\operatorname{Re}(z) = \text{real part of } z \\ = 2$$

$$\operatorname{Im}(z) = \text{imaginary part of } z \\ = +3i$$

$$\text{conjugate of "z"} = z^* = \bar{z}$$

$$\text{conjugate of } 2 + 3i \text{ is } 2 - 3i$$

e.g. $z = 7 + 4i$ $w = 2 - 3i$

$$(a) \operatorname{Re}(z) = 7$$

$$(b) \operatorname{Im}(w) = -3i$$

$$(c) w^* = 2 + 3i$$

$$(d) \operatorname{Re}(w) + \operatorname{Im}(z) = 2 + 4i$$

Arithmetic $(+, -, \times, \div)$

Mode 2

ENG : i

$$z = 7 + 4i$$

$$w = 2 - 3i$$

$$\begin{aligned} z + w &= (7 + 2) + (4i - 3i) \\ &= 9 + i \end{aligned}$$

$$\begin{aligned} z - w &= (7 - 2) + (4i - (-3i)) \\ &= 5 + 7i \end{aligned}$$

$$\begin{aligned} z \times w &= (7 + 4i)(2 - 3i) \\ &= 14 - 21i + 8i - 12i^2 \\ &= 14 - 13i - 12(-1) = 14 - 13i + 12 \\ &= 26 - 13i \end{aligned}$$

$$i^2 = -1$$

$$\begin{aligned} z \times z^* &= (7 + 4i)(7 - 4i) \\ &= 7^2 - (4i)^2 = 49 - 16i^2 = 49 + 16 \\ &= 65 \end{aligned}$$

(Always give real no., no imaginary part)

★ NOTE: A complex number multiplied by its conjugate always gives a REAL NUMBER.

Division

$$\frac{z}{w} = \frac{z}{w} \times \frac{w^*}{w^*} = \frac{zw^*}{ww^*}$$

$$\begin{aligned} \frac{z}{w} &= \frac{7 + 4i}{2 - 3i} \times \frac{(2 + 3i)}{(2 + 3i)} = \frac{(7 + 4i)(2 + 3i)}{(2 - 3i)(2 + 3i)} \\ &= \frac{14 + 21i + 8i + 12i^2}{4 - 9i^2} = \frac{14 + 29i - 12}{4 + 9} = \frac{2 + 29i}{13} \\ &= \frac{2}{13} + \frac{29}{13}i \end{aligned}$$

★ NOTE: All real numbers are complex numbers $7 + 0i$
Not All complex numbers are real numbers. $2 + 3i$ (not real)
 $0 + 7i$

SOLVE THE EQUATION

$$4x^2 - 4x + 17 = 0$$

$$\begin{aligned} x &= \frac{4 \pm \sqrt{(-4)^2 - 4(4)(17)}}{8} \\ &= \frac{4 \pm \sqrt{-256}}{8} = \frac{4 \pm \sqrt{256}i}{8} \\ &= \frac{4 \pm 16i}{8} = \frac{1}{2} \pm 2i \end{aligned}$$

FINDING THE SQUARE ROOTS of a CN

E.g. Find the 2 square roots of $8 + 6i$

$$8 + 6i = \boxed{?}^2$$

$$\begin{aligned} \text{Let } \underbrace{8}_{\text{Re}} + \underbrace{6i}_{\text{Im}} &= (a + bi)^2 \quad \text{where } a \text{ and } b \text{ are real} \\ &= (a + bi)(a + bi) \\ &= a^2 + abi + abi + b^2i^2 \\ &= a^2 + 2abi - b^2 \\ &= \underbrace{(a^2 - b^2)}_{\text{Re}} + \underbrace{2abi}_{\text{Im}} \end{aligned}$$

$$a^2 - b^2 = 8$$

$$a^2 - \left(\frac{3}{a}\right)^2 = 8$$

$$a^2 - \frac{9}{a^2} = 8$$

$$a^4 - 9 = 8a^2$$

$$(a^2)^2 - 8a^2 - 9 = 0 \quad \therefore \text{SO, roots are } a + bi$$

$$(a^2 - 9)(a^2 + 1) = 0 \quad \therefore 3 + i \text{ or } -3 - i$$

$$a^2 = 9 \quad \text{or} \quad a^2 = -1$$

$$a = \pm 3 \quad a = \sqrt{-1} = i$$

Not valid / a is real number

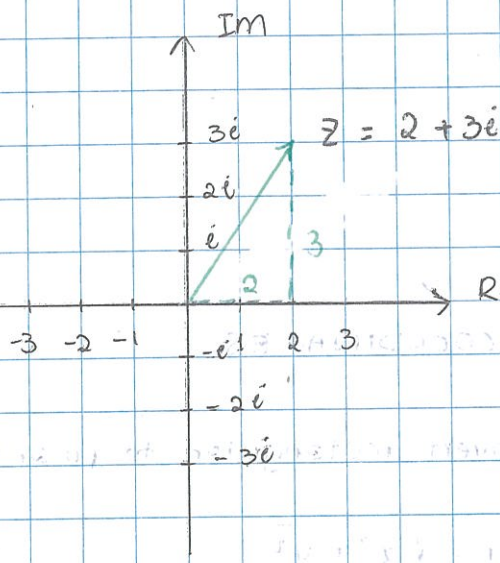
$$2ab = 6$$

$$b = \frac{3}{a}$$

$$\text{and } b = \frac{3}{a}$$

$$b = 1 \quad \text{or} \quad -1$$

ARGAND DIAGRAM (complex number planes)



each complex number is a point on the 2-D diagram

real part = horizontal coordinate

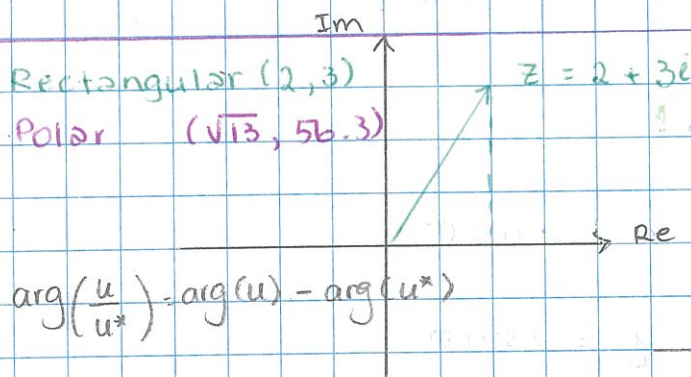
Imaginary part = vertical "

Sometimes represented as vectors

$$z = 2 + 3i$$

can be added / subtracted like vectors

THE MODULUS and ARGUMENT of a CN



Rectangular (2, 3)

Polar $(\sqrt{13}, 56.3)$

$$z = 2 + 3i$$

Modulus of $z = |z|$ [mod(z)]
= distance from (0,0)

$$0 \leq |z|$$

argument of $z = \arg(z)$

= angle of rotation from positive x-axis.

Positive
← + anti-clockwise
↻ - clockwise

$$-180^\circ < \arg(z) < 180^\circ$$

E.g. Finding the modulus and argument of:

(a) $2 + 3i$

$$|z| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

$$\tan \theta = \frac{3}{2}$$

$$\theta = \tan^{-1}\left(\frac{3}{2}\right) = 56.30993\dots$$

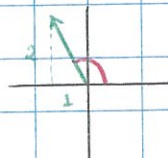
$$= 56.3$$

The answer usually in degree

$$\arg(z^*) = -56.3$$

(b) $-1 + 2i$ ignore the (-) because |z| will always be (+)

$$|z| = \sqrt{1^2 + 2^2} = \sqrt{5}$$



$$\arg(z) = 180^\circ - \tan^{-1}(2)$$

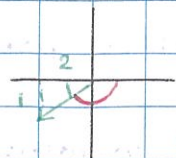
$$= 180^\circ - 63.43494\dots$$

$$= 116.5650\dots$$

$$= 116.6^\circ$$

$$(c) -2 - i$$

$$|z| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

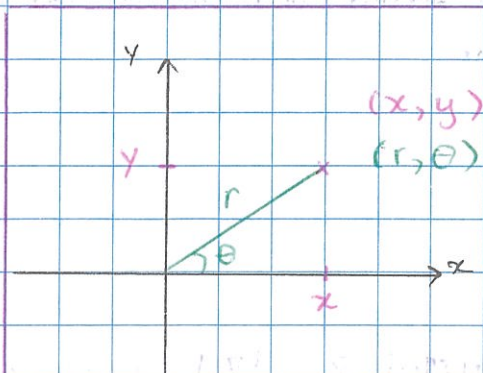


$$\arg(z) = -\left(180^\circ - \tan^{-1}\left(\frac{1}{2}\right)\right)$$

$$= -153.4349^\circ$$

$$= -153.4^\circ$$

RECTANGULAR VS. POLAR COORDINATES



convert rectangular to polar:

$$r = \sqrt{x^2 + y^2}$$

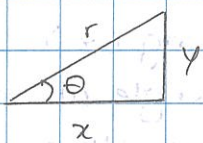
$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Also, $\text{Pol}\left(\frac{x}{r}, \frac{y}{r}\right)$

key in the
calculator

$$r = \dots \quad \theta = \dots$$

convert polar to rectangular:



$$\cos \theta = \frac{x}{r}$$

$$x = r \cos \theta$$

Also, $\text{Rec}(r, \theta)$

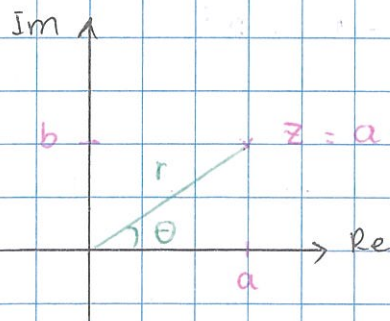
key

$$\sin \theta = \frac{y}{r}$$

$$y = r \sin \theta$$

$$x = \dots \quad y = \dots$$

COMPLEX NUMBER in POLAR FORM



$$z = a + bi$$

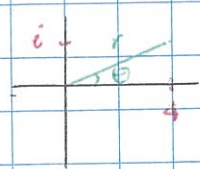
$$= (r \cos \theta) + (r \sin \theta) i$$

$$= r (\cos \theta + i \sin \theta)$$

↑
CN in Polar form

E.g. write $z = 4 + i$ in polar form.

→ shift 2 → 3 → $\sqrt{17} < 14.0^\circ$



$$r = \sqrt{4^2 + 1^2}$$

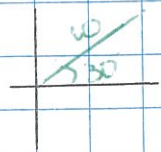
$$r = \sqrt{17}$$

$$\theta = \tan^{-1}\left(\frac{1}{4}\right) = 14.0^\circ$$

$$z = \sqrt{17} (\cos 14.0^\circ + i \sin 14.0^\circ) //$$

E.g. $z = 10 (\cos 30^\circ + i \sin 30^\circ)$ if -100°

write z in rectangular (cartesian) form



$$z = 10 \cos 30^\circ + (10 \sin 30^\circ)i$$

$$= 10 \frac{\sqrt{3}}{2} + 10 \times \frac{1}{2} i$$

$$= 5\sqrt{3} + 5i //$$

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2$$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$

• rectangular (x, y) $5\sqrt{3} + 5i$

• polar (r, θ) $(10, 30^\circ)$

X, ÷ CNS in POLAR FORM

If $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$
and

$$z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$$

$$\text{then } z_1 \times z_2 = r_1 \times r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$$

"multiply moduli"

"add arguments"

$$\text{and } \frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2))$$

"Divide moduli"

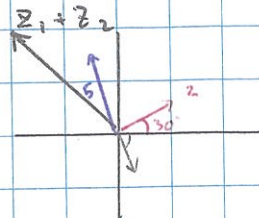
"subtract arguments"

$$\text{e.g. } z_1 = 2 (\cos 30^\circ + i \sin 30^\circ) \quad z_2 = 5 (\cos 100^\circ + i \sin 100^\circ)$$

$$z_1 \times z_2 = 10 (\cos 130^\circ + i \sin 130^\circ)$$

$$\frac{z_1}{z_2} = \frac{2}{5} (\cos -70^\circ + i \sin -70^\circ)$$

$$z_1^3 = z_1 \times z_1 \times z_1 = 8 (\cos 90^\circ + i \sin 90^\circ)$$



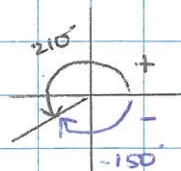
POWERS OF CNS

$$\text{if } Z = r(\cos \theta + i \sin \theta)$$

$$\text{then, } Z^n = r^n (\cos(n\theta) + i \sin(n\theta))$$

$$\text{e.g. } Z = 3(\cos 70^\circ + i \sin 70^\circ)$$

$$\begin{aligned} Z^3 &= 3^3 (\cos(3 \times 70^\circ) + i \sin(3 \times 70^\circ)) \\ &= 27 (\cos 210^\circ + i \sin 210^\circ) \\ &= 27 (\cos -150^\circ + i \sin -150^\circ) \end{aligned}$$



SQUARE ROOTS ($Z^{\frac{1}{2}}$)

Always come in pairs

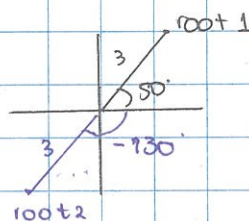
$$\text{First root} = r^{\frac{1}{2}} (\cos(\frac{1}{2}\theta) + i \sin(\frac{1}{2}\theta))$$

Second root will be diagonally opposite first root

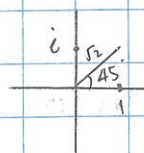
$$Z = 9(\cos 100^\circ + i \sin 100^\circ)$$

$$\text{first root} = 3(\cos 50^\circ + i \sin 50^\circ)$$

$$\text{second root} = 3(\cos -130^\circ + i \sin -130^\circ)$$



Find 2 square root of $1 + i$

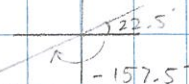


$$\sqrt{2} (\cos 45^\circ + i \sin 45^\circ)$$

$$\text{root 1} = \sqrt[4]{2} (\cos 22.5^\circ + i \sin 22.5^\circ)$$

$$= \sqrt[4]{2} (\cos 22.5^\circ + i \sin 22.5^\circ)$$

$$\text{root 2} = \sqrt[4]{2} (\cos -157.5^\circ + i \sin -157.5^\circ)$$



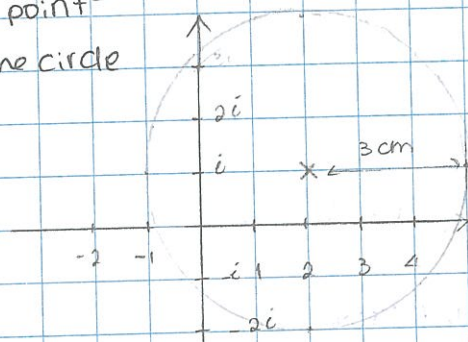
COMPLEX LOCI

Loci = Set of points that satisfy a geometric condition

Loci of the form $|z - z_1| = C$ (C - constant)
 $\uparrow$ must be - (distance) z_1 is a given point

will be a circle, centre = z_1 , radius = C

e.g. $|z - (2 + i)| = 3$
any point $\swarrow$
on the circle



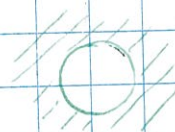
NOTE: If " ≤ 3 "



" < 3 "



" ≥ 3 "



7.8.20

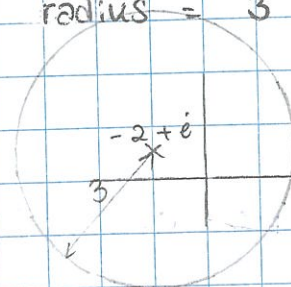
NOTE: It must be written like this:
 $|z - (a + bi)| = \text{constant}$

e.g. $|z + 2 - i| = 3$

rewrite $\rightarrow |z - (-2 + i)| = 3$

centre = $(-2 + i)$

radius = 3



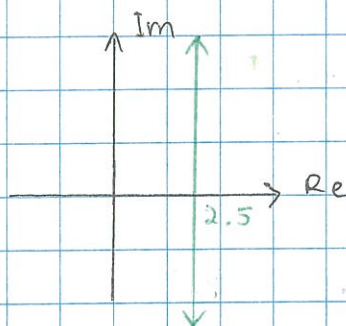
NOTE: If centre is $0 + 0i$,
the equation will be
 $|z| = \text{constant}$

$\uparrow$
 $|z - (0 + 0i)| = 3$

LOCI OF THE FORM

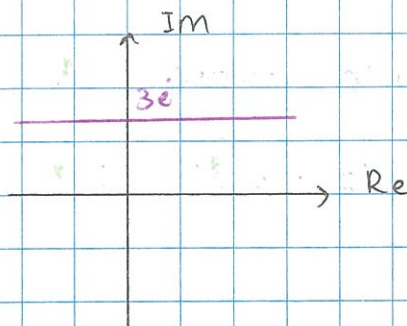
$$\operatorname{Re}(z) = \text{constant}$$

or $\operatorname{Im}(z) = \text{constant}$



$$\operatorname{Re}(z) = 2.5$$

(vertical line)

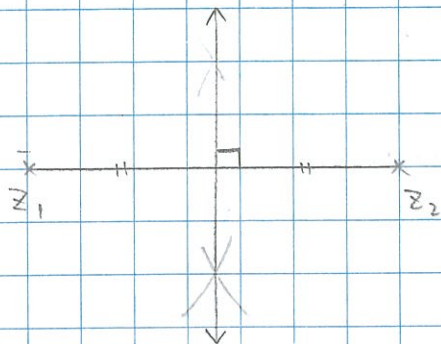


$$\operatorname{Im}(z) = 3i$$

(Horizontal line)

LOCI INVOLVING PERPENDICULAR BISECTOR

$$|z - z_1| = |z - z_2| \rightarrow \text{perpendicular bisector through } z_1 \text{ and } z_2$$

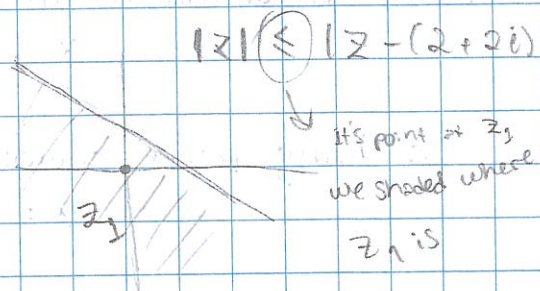
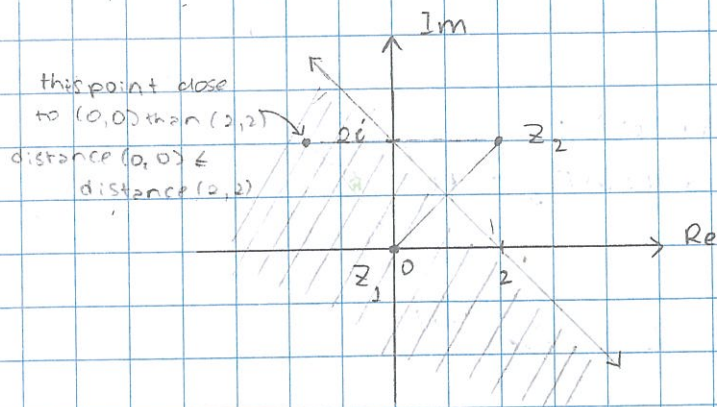


NOTE: If $z_1 = 0 + 0i$: $|z| = |z - z_1|$

e.g sketch $|z| \leq |z - (2 + 2i)|$

$$|z - (0 + 0i)| \leq |z - (2 + 2i)| \rightarrow \text{distance to } (0,0) \leq \text{distance to } (2,2)$$

z_1 z_2

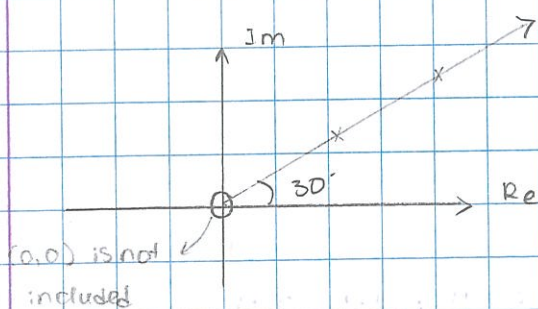


LOCI INVOLVING ARGUMENT OF z $\arg(z)$

$\arg(z) = \text{constant}$

$\arg$ = angle with + horizontal

e.g $\arg(z) = 30^\circ$

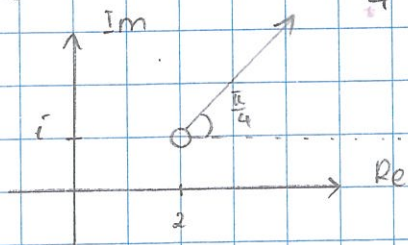


The locus is a 'ray' starting from (0,0)

$\arg(z - z_1) = \text{constant}$

↑ must be '-'

$$\arg(z - (2 + i)) = \frac{\pi}{4}$$



The locus is a ray starting at z_1

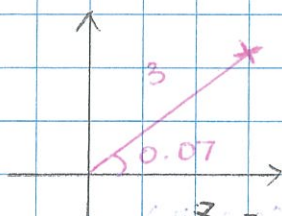
THE FORM $Z = re^{i\theta}$

If Z is a complex number where $|Z| = r$ and $\arg(Z) = \theta$, then Z can be written in the form:

$$Z = re^{i\theta} \quad [\theta \text{ in radians}]$$

So, using polar coordinates (r, θ) :

$$Z = r(\cos \theta + i \sin \theta) = re^{i\theta}$$



$$Z = 3e^{0.7i}$$

$$Z = 3(\cos 0.7 + i \sin 0.7)$$

Examples:

Find (i) (a) $4e^{5i} \times 3e^{2i} = 12e^{7i}$

time the modulus $4 \times 3 = 12$

Add the argument $5i + 2i = 7i$

(b) $6e^{9i} \div 3e^{2i} = 2e^{7i}$

Divide the modulus $6 \div 3 = 2$

Subtract the angles $9i - 2i = 7i$

in radians

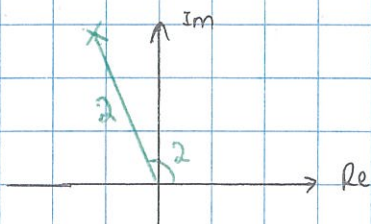
(ii) write these result as complex numbers in polar form

(a) $4(\cos 5 + i \sin 5) \times 3(\cos 2 + i \sin 2) = 12(\cos 7 + i \sin 7)$

(b) $6(\cos 9 + i \sin 9) \div 3(\cos 2 + i \sin 2) = 2(\cos 7 + i \sin 7)$

Ex - Draw $Z = 2e^{2i}$ on a Argand diagram

$|Z| = 2$ $\arg(Z) = 2 \text{ radian} = \frac{2}{\pi} \times 180^\circ = 114.6^\circ$



COMPLEX NUMBER & SOLUTION OF POLYNOMIALS

The conjugate root theorem

Any polynomial which has real coefficients only, has roots in conjugate pairs. i.e. if α is a root then α^* is also a root

$$-2 + i$$

$$-2 - i \rightarrow \bar{\alpha}$$

Example 1: Let $f(z) = z^3 - 12z^2 + 46z - 52 \rightarrow$ real coefficient
given that $5+i$ is a root of $f(z)$, solve equation $f(z) = 0$

$$\alpha_1 = 5 + i$$

$$\alpha_2 = 5 - i$$

$$\alpha_3 = ?$$

$z-5$
will give ans 5

$$z^3 - 12z^2 + 46z - 52 = (z - \alpha_1)(z - \alpha_2)(z - \alpha_3) \\ = (z - (5+i))(z - (5-i))(z - \alpha_3)$$

$$\text{So, } -(5+i)x - (5-i)x - \alpha_3 = -52$$

$$(5+i)(5-i)x - \alpha_3 = 52$$

$$(25 - i^2)x - \alpha_3 = 52$$

$$26x - \alpha_3 = 52$$

$$x = \frac{52}{26} = 2$$

So, factors of $f(z)$ are $(z - (5+i))$
(brackets) $(z - (5-i))$

$$(z - 2)$$

→ check the ans on calculator
mode 5

Roots of $f(z)$ are $5+i, 5-i, 2$

Example 2 Let $f(z) = z^4 + 4$

Show that $1+i$ is a root of $f(z)$ and find all solutions

to the equation $f(z) = 0$ $\rightarrow$ it means show $f(1+i) = 0$

$$\text{Let } z = 1+i$$

$$z^2 = (1+i)^2 = 1 + 2i + i^2 = 2i$$

$$z^4 = (z^2)^2 = (2i)^2 = 4i^2 = -4$$

$$\begin{aligned}\text{So, } f(1+i) &= (1+i)^4 + 4 \\ &= -4 + 4 \\ &= 0\end{aligned}$$

$$\alpha_1 = 1+i \quad \alpha_2 = 1-i \quad \alpha_3 = ? \quad \alpha_4 = ?$$

$$\begin{aligned}\text{Now } f(z) &= (z - (1+i))(z - (1-i))(z^2 + az + b) \\ &= [z^2 - (1-i)z - (1+i)z + (1+i)(1-i)](z^2 + az + b) \\ &= (z^2 - 2z + 1 - i^2)(z^2 + az + b) \\ z^4 + 4 &= (z^2 - 2z + 2)(z^2 + az + b) \\ 2b &= 4 \rightarrow b = 2\end{aligned}$$

$$0z = -2bz + 2az$$

$$0z = -4z + 2az$$

$$\text{So, } a = 2$$

$$\text{So, } (z^2 - 2z + 2)(\underbrace{z^2 + 2z + 2}_{2 \text{ roots}})$$

$$\begin{aligned}z &= \frac{-2 \pm \sqrt{4 - 4 \times 1 \times 2}}{2} = \frac{-2 \pm \sqrt{4}}{2} \\ &= \frac{-2 \pm 2i}{2} = -1 + i \text{ or } -1 - i\end{aligned}$$

$$4 \text{ roots : } -1+i, -1-i, 1+i, 1-i$$